

REDUCTIONS AND PARTIAL RESULTS TOWARDS THE DVOŘÁK–MOHAR–ŠÁMAL STAR 6-EDGE-COLOURING CONJECTURE

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ABSTRACT. Dvořák, Mohar and Šámal conjectured that every subcubic multigraph has star chromatic index at most 6; the best published general bound is 7. We do *not* prove the conjecture. We report a chain of reductions, obtained with the help of an automated proof-search system and checked by its verification pipeline, which shows that the conjecture (for finite loopless multigraphs) follows from five explicit statements about 2-cut-reduced cubic multigraphs: two “far-exchange” statements on digon insertions of cyclically 4-edge-connected simple cubic graphs, a pole-dominance statement, a restricted perfect-matching colouring property at triangles carrying one digon, and a leaf-attachment statement. The assembly of this chain, together with all finite small-case facts except three census-level hypotheses, is formalized in *Lean* with only the standard axioms. We also record *Lean*-certified finite base cases (all 2-cut-reduced cubic multigraphs on 10 and 12 vertices, and the digon insertions of small cyclically 4-edge-connected hosts), partial results on each open statement, star colourings of several infinite families (generalized Petersen graphs $GP(k, 2)$ with the spokes as a colour class, flower and Goldberg snarks, covers of the Petersen graph and of K_4), and an explicit account of which parts are machine-checked and which rest on model-based verification only.

1. INTRODUCTION

A *star edge colouring* of a multigraph is a proper edge colouring with no bicoloured path or cycle of length four. The *star chromatic index* $\chi'_s(G)$ is the least number of colours in such a colouring. Dvořák, Mohar and Šámal [1] proved that every subcubic multigraph satisfies $\chi'_s(G) \leq 7$ and conjectured the following.

Conjecture ((DMS), [1]). *Every subcubic multigraph has star chromatic index at most 6.*

The bound 6 would be sharp ($K_{3,3}$ needs six colours [1]). Partial results include the bound 6 for subcubic multigraphs of maximum average degree below $5/2$ and the bound 5 below $24/11$ (Lei, Shi and Song [2]; see also [3]), and bounds for special classes such as Halin graphs and generalized Petersen graphs [4, 5, 6]. The conjecture remains open.

What this paper contains. This is a *partial-results* report. Its mathematical content was produced by an automated multi-agent proof-search system (a fork of the *Danus* harness) working on the conjecture for *finite loopless* multigraphs, and every result stated here is a fact that passed the system’s verification pipeline (Section 9 explains exactly what that means, result by result). Our contributions are:

- (1) **A reduction theorem** (Main Theorem 3.9): DMS for loopless multigraphs follows from five explicitly stated statements $(\text{FE-EXT-D})_{10}$, $(\text{FE-EXIST-D})_{10}$, (POLE) , (TD-TRI) , (II_D) about 2-cut-reduced cubic multigraphs on at least 10 (in one case 6) vertices. Along the way we record several equivalent or sufficient single-statement forms (Theorems 3.1–3.3).
- (2) **A formal version** (Theorem 3.12): the whole reduction, including all its finite inputs except three census-level finite hypotheses, is proved in *Lean* with standard axioms only.
- (3) **Finite certified cases** (Section 4): certificates for all small cases used, most of them kernel-checked in *Lean*.

- (4) **Partial progress on each open statement** (Section 5), e.g. (POLE) holds up to 16 vertices and (Π_D) up to 14 vertices.
- (5) **Class results** (Section 6), with a comparison to the literature.
- (6) **Open problems and obstacles** (Section 8) and an honest **verification status** (Section 9).

Caveats up front. None of the five hypotheses of Main Theorem 3.9 is proved; each is supported by computation only in the ranges stated in Sections 5 and 7. Most of the facts were checked by a language-model verifier and a second-model cross-check audit rather than by a proof assistant; several related facts carry open audit disputes (Section 9). The novelty of most results has not been checked against the literature (Section 9.4).

2. DEFINITIONS AND CONVENTIONS

All multigraphs are finite and loopless; parallel edges are allowed and are distinct edges. Two distinct edges *meet* if they share an endpoint (parallel edges meet). A multigraph is *cubic* (resp. *subcubic*) if every vertex is incident with exactly (resp. at most) three edges. For $S \subseteq V(X)$, $\delta(S)$ is the set of edges with exactly one endpoint in S ; S is a *k-edge-cut side* if $\emptyset \neq S \neq V(X)$ and $|\delta(S)| = k$. A *perfect matching* (PM) is a set of pairwise vertex-disjoint edges covering every vertex. For an edge g and an edge set N we write $[g \in N] \in \{0, 1\}$.

Definition 2.1 (Star colourings). A *star edge colouring with k colours* of X is a map $c: E(X) \rightarrow \{1, \dots, k\}$ such that (i) distinct edges with a common endpoint get different colours, and (ii) X has no path with exactly four edges (on five distinct vertices) and no cycle with exactly four edges (on four distinct vertices) whose consecutive edges e_1, e_2, e_3, e_4 satisfy $c(e_1) = c(e_3)$ and $c(e_2) = c(e_4)$. An edge set N is a *colour class* of c if $N = c^{-1}(\mu)$ for some μ . An *MC colouring* is a star edge colouring with 6 colours whose colour class 6 is a perfect matching; (X, N) is *MC-colourable* if X has an MC colouring with $c^{-1}(6) = N$.

Definition 2.2 (Classes). $\mathcal{G}$ is the class of connected bridgeless loopless cubic multigraphs. A member X of $\mathcal{G}$ is *2-cut-reduced* if every 2-edge-cut side S satisfies $|S| = 2$ or $|V(X) \setminus S| = 2$; $\mathcal{D} \subseteq \mathcal{G}$ is the class of 2-cut-reduced members. $\mathcal{S}$ is the class of simple 3-edge-connected cubic graphs. A *cyclic 3-edge-cut side* of a cubic X is a 3-edge-cut side S such that both $X[S]$ and $X[V(X) \setminus S]$ contain a cycle; a cubic graph is *cyclically 4-edge-connected* (c4c) if it is connected and, for every set R of at most three edges, at most one component of $X - R$ contains a cycle. For $H \in \mathcal{S}$, $s(H)$ is the least size of a cyclic 3-edge-cut side of H .

Definition 2.3 (Digon insertion). For a cubic graph H and $D \subseteq E(H)$, the *digon insertion* H^D replaces every edge $ab \in D$ by two new vertices u_{ab}, v_{ab} , the edges $au_{ab}, v_{ab}b$, and two parallel edges $u_{ab}v_{ab}$. Thus $|V(H^D)| = |V(H)| + 2|D|$.

Definition 2.4 (Leaf graph). For a loopless cubic multigraph G_0 and an edge g of G_0 with ends $s \neq t$, the *leaf graph* $T(G_0, g)$ is obtained from G_0 by deleting g and adding two new vertices x, ℓ and the three edges $sx, xt, x\ell$. An edge g of X is *eligible* if it has distinct ends s, t , no other edge joins s and t , and g lies in no 2-edge-cut of X .

Definition 2.5 (EX1-good). A connected loopless cubic multigraph X is *EX1-good* if for every edge g and every $t \in \{0, 1\}$ such that some PM N of X has $[g \in N] = t$, there is a PM N of X with $[g \in N] = t$ such that (X, N) is MC-colourable.

Definition 2.6 (Far-exchange sets). Let X be a loopless cubic multigraph and P a PM of X . The *distance* between two edges is the least number of edges of a path from an endpoint of one to an endpoint of the other. A set $C \subseteq E(X) \setminus P$ is a *far-exchange set for P* if

- (a) every edge of P meets at most one edge of C ;

- (b') no edge of $E(X) \setminus (P \cup C)$ that is parallel to no edge of C has both endpoints incident with edges of C ;
- (c) any two distinct edges of P that meet no edge of C are at distance at least 3.

Definition 2.7 (Poles and dominance). Let X be a loopless cubic multigraph and v a vertex whose three edges join it to three distinct vertices x_1, x_2, x_3 . The *pole* $Q(X, v)$ is $X - v$ together with three new vertices o_t and *port edges* $p_t = x_t o_t$ ($t = 1, 2, 3$). For $d: E(Q) \rightarrow \{1, \dots, 6\}$, $\text{Col}_t(d)$ is the set of colours of the two edges $f \neq p_t$ at x_t , and $\text{Blk}_t(d)$ is the set of those colours $d(f)$, $f = x_t r$, for which r is incident with an edge $f' \neq f$ with $d(f') = d(p_t)$. The map d is an *MC pole colouring* if it is a star edge colouring with 6 colours and every vertex of $X - v$ is incident with exactly one edge of colour 6.

Let $i \in \{1, 2, 3\}$, $\{j, k\} = \{1, 2, 3\} \setminus \{i\}$. An *admissible outside datum* at M -port i consists of pairwise distinct colours a_1, a_2, a_3 with $a_i = 6$ and sets $B_t \subseteq O_t \subseteq \{1, \dots, 6\}$ with $|O_t| = 2$ such that: $O_i \subseteq \{1, \dots, 5\} \setminus \{a_j, a_k\}$ and $B_i = O_i$; for $t \in \{j, k\}$, $6 \in O_t$, $a_t \notin O_t$ and $B_t \subseteq O_t \setminus \{6, a_j, a_k\}$; and not both $a_k \in O_j$ and $a_j \in O_k$. An MC pole colouring d is *compatible* with the datum if $d(p_t) = a_t$ and $\text{Col}_t(d) \cap O_t \cap (\text{Blk}_t(d) \cup B_t) = \emptyset$ for $t = 1, 2, 3$. The pole is *dominant* if

- (D1) for every i and every admissible outside datum at M -port i there is a compatible MC pole colouring; and
- (D2) for every edge g of $X - v$ and $s \in \{0, 1\}$ there is i such that for every admissible outside datum at M -port i there is a compatible MC pole colouring d with $[d(g) = 6] = s$.

Definition 2.8 (The property (TD)). Let $X \in \mathcal{G}$ and let y_1, y_2, y_3, u, v be distinct vertices such that the edges incident with them are exactly $y_1 y_2, y_1 y_3, y_2 u$, two parallel edges uv, vy_3 , and three edges $e_t = y_t w_t$ with w_1, w_2, w_3 distinct and outside $\{y_1, y_2, y_3, u, v\}$ (a triangle carrying one digon); let $U = V(X) \setminus \{y_1, y_2, y_3, u, v\}$. (TD) holds for $(X; y_1, y_2, y_3, u, v)$ if for every edge g of $X[U]$ and $t \in \{0, 1\}$ such that some PM $N \ni e_1, y_2 u, vy_3$ has $[g \in N] = t$, there is an MC colouring c of X with $c(e_1) = c(y_2 u) = c(vy_3) = 6$ and $[c(g) = 6] = t$.

3. THE REDUCTION CHAIN

3.1. From all subcubic multigraphs to two cubic statements. Consider

- (I) every bridgeless loopless cubic multigraph has a star edge colouring with 6 colours;
- (II) for every bridgeless loopless cubic multigraph G_0 and every edge g of G_0 , $T(G_0, g)$ has a star edge colouring with 6 colours.

Theorem 3.1. (I) $\wedge$ (II) holds if and only if every finite loopless multigraph of maximum degree at most 3 has a star edge colouring with 6 colours.

Proof outline. The converse is trivial. For the forward direction one argues by induction on $|E(H)|$. Components are coloured separately. A bridge with edges on both sides is handled by a gluing lemma for star 6-colourings across a bridge. Otherwise every bridge is pendant; the core C obtained by deleting the leaves is connected, bridgeless, with degrees 2 and 3. If C has no degree-2 vertex, $H = C$ and (I) applies. If it has $k \geq 2$ of them, H embeds into a bridgeless cubic multigraph obtained by joining the (existing or new) leaves in a cycle (or a digon if $k = 2$), and (I) applies to it. If $k = 1$, suppressing the degree-2 vertex y of C yields a bridgeless cubic G_0 with a new edge g , and H is a sub-multigraph of $T(G_0, g)$, so (II) applies. Star colourings restrict to sub-multigraphs. $\square$

3.2. Single-statement sufficient conditions. Two strengthenings give DMS on their own; both implications are formalized.

Theorem 3.2 (CubicSharp5_s implies DMS). Suppose every connected bridgeless simple cubic graph other than $K_{3,3}$ and the prism $K_3 \square K_2$ has a star edge colouring with 5 colours. Then (I) and (II) hold, and hence every finite loopless multigraph with maximum degree at most 3 has a star edge colouring with 6 colours.

The proof replaces each digon of a cubic multigraph by a rigid 5-colourable gadget (the prism minus an edge), colours the resulting simple graph with 5 colours, and uses the sixth colour on one edge of each digon; the leaf case is a finite case analysis on the position of g relative to the digons, which is itself formalized.

Theorem 3.3 (HOLE implies DMS). *Suppose that for every $X \in \mathcal{G}$ not isomorphic to $K_{3,3}$ and every vertex t there are a star edge colouring c of X with 6 colours and a colour β missing from all edges incident with the closed neighbourhood $N[t]$. Then every finite loopless multigraph with maximum degree at most 3 has a star edge colouring with 6 colours.*

Remark 3.4. The hypotheses of Theorems 3.2 and 3.3 are open. For CubicSharp5_s the only evidence is computational (Section 7); we did not pursue these routes further.

3.3. Perfect matchings as a colour class. The main route replaces (I) by a statement asking for a perfect matching as a colour class.

- (H) every $X \in \mathcal{D}$ with $|V(X)| \geq 10$ is EX1-good.
- (II_D) for every $G_0 \in \mathcal{D}$ with $|V(G_0)| \geq 6$ and every eligible edge g of G_0 , $T(G_0, g)$ has a star edge colouring with 6 colours.

Theorem 3.5 (EX1 reduction). *If (H) holds, then every $X \in \mathcal{G}$ with $|V(X)| \geq 10$ is EX1-good; in particular it has a PM M and a star 6-edge-colouring with M as a colour class.*

Theorem 3.6 (Reduction to (H) and the leaf case). *(a) (H) $\wedge$ (II) implies DMS for loopless multigraphs.*

(b) (H) $\wedge$ (II_D) implies (II); hence (H) $\wedge$ (II_D) implies DMS for loopless multigraphs.

Both statements are proved in Lean (Lean 4.20 with Mathlib v4.20.0, standard axioms only), and the Lean propositions for (H), (II), (II_D) and DMS are proved equivalent, in Lean, to plain restatements of the prose over arbitrary finite types.

Proof outline. (a) Components with at most 8 vertices are handled by a finite base case and larger ones by Theorem 3.5; this gives (I), and Theorem 3.1 finishes. (b) Under (H), the leaf case reduces across 2-edge-cuts and digons to 2-cut-reduced hosts at eligible edges, using gluing lemmas for MC colourings across 2-edge-cuts. $\square$

3.4. From (H) to statements about c4c hosts. Every member of $\mathcal{D}$ on at least 10 vertices is a digon insertion H^D of some $H \in \mathcal{S}$. The hypothesis (H) is split along this structure: the c4c hosts are handled by two “far-exchange” statements, cyclic 3-edge-cuts by a gluing (“brick”) lemma that needs pole dominance of one side, and the one configuration that the brick does not cover—a triangle side of minimum size carrying exactly one digon—by the property (TD).

Statement 3.7 (The five open statements).

- (FE-EXT-D)₁₀ For every c4c simple cubic graph H with $|V(H)| \geq 10$, every $D \subseteq E(H)$ with $|V(H)| + 2|D| \geq 16$, every PM P of $X := H^D$ and every far-exchange set C for P in X , X has a star edge colouring c with 6 colours such that $c^{-1}(6) = P$ and $c^{-1}(5) = C$.
- (FE-EXIST-D)₁₀ For every c4c simple cubic graph H with $|V(H)| \geq 10$, every $D \subseteq E(H)$ with $|V(H)| + 2|D| \geq 16$, every edge g of $X := H^D$ and every $t \in \{0, 1\}$, there are a PM P of X with $[g \in P] = t$ and a far-exchange set C for P in X .
- (POLE) For every $X \in \mathcal{D}$ with $|V(X)| \geq 10$ and every vertex v of X whose three edges join it to three distinct vertices, the pole $Q(X, v)$ is dominant.
- (TD-TRI) For every $H \in \mathcal{S}$ that is not c4c, every cyclic 3-edge-cut side $S = \{y_1, y_2, y_3\}$ with $|S| = 3 = s(H)$, and every $D \subseteq E(H)$ with $|V(H)| + 2|D| \geq 16$ such that the edges of D incident with S consist of exactly one edge y_2y_3 (after renaming), (TD) holds for $(H^D; y_1, y_2, y_3, u, v)$, where u, v are the new vertices of the digon inserted into y_2y_3 (u adjacent to y_2).

(II_D) As above: for every $G_0 \in \mathcal{D}$ with $|V(G_0)| \geq 6$ and every eligible edge g , $T(G_0, g)$ is star 6-edge-colourable.

Theorem 3.8 (Intermediate reductions). *(a) $(\text{FE-EXT-D})_{10} \wedge (\text{FE-EXIST-D})_{10}$ implies: for every c4c simple cubic H with $|V(H)| \geq 10$ and every D with $|V(H)| + 2|D| \geq 16$, H^D is EX1-good.*

(b) For c4c simple cubic hosts on at most 8 vertices the same conclusion holds unconditionally (Lemma SMALLHOST-D, Section 4).

(c) The conclusion of (a), (POLE) and (TD-TRI), together with the finite facts of Section 4, imply (H).

Proof outline. (a) Given g, t , (FE-EXIST-D)₁₀ provides (P, C) with $[g \in P] = t$, and (FE-EXT-D)₁₀ colours H^D with P as colour class 6. (c) By induction over cyclic 3-edge-cuts of the host: a 3-edge-cut with distinct ends whose one contraction has a dominant pole and whose other contraction is EX1-good yields EX1-goodness of the whole graph (the “3-cut brick”); (POLE) supplies dominance. At a minimum triangle side carrying exactly one digon the pole is not available in the needed form, and (TD) is used instead (Lemma TRI-BRICK-TD). The finite facts cover the hosts with 14 vertices and the small contractions. $\square$

3.5. The main reduction theorem.

Theorem 3.9 (Main reduction; ROOT-CS4). *If $(\text{FE-EXT-D})_{10}$, $(\text{FE-EXIST-D})_{10}$, (POLE), (TD-TRI) and (II_D) hold, then (H) and (II) hold, and every finite loopless multigraph with maximum degree at most 3 has a star edge colouring with 6 colours.*

Proof. Theorem 3.8(a),(b),(c) give (H); Theorem 3.6(b) gives (II) and DMS. $\square$

Remark 3.10 (Variants). The same conclusion holds with (TD-TRI) replaced by an “L-shape” form (TD-L-TRI) (which implies it), and with (POLE) replaced by a single boundary-extension statement (W-EXT) (“one universal W-colouring per M-port”, which implies (POLE)). Moreover (POLE) is equivalent to its restriction to members with at least 18 vertices, split into those with no 3-edge-cut with distinct ends and an explicitly described residue.

Remark 3.11 (Scope of the hypotheses). Only the two far-exchange statements are about c4c hosts. (POLE) and (II_D) range over all of $\mathcal{D}$ (above 10 resp. 6 vertices), and (TD-TRI) over non-c4c hosts with a minimum triangle side. The restriction to c4c hosts in the far-exchange statements is essential: the analogous existence statement over all of $\mathcal{D}$ is false, e.g. for an 18-vertex simple 3-edge-connected cubic graph (a modification of $K_{3,3}$) and an edge g such that none of the 12 PMs containing g admits any far-exchange set.

3.6. The formal version.

Theorem 3.12 (Lean). *In the project’s Lean library (Lean 4.20 with Mathlib v4.20.0; only the axioms `propext`, `Classical.choice`, `Quot.sound`; no `sorry`, no `native_decide`) the following are proved:*

- (1) *SMALLHOSTD (Lemma SMALLHOST-D) unconditionally;*
- (2) *BASE12 $\rightarrow$ SIMPLE14 $\rightarrow$ B14D $\rightarrow$ FEEXTD10 $\rightarrow$ FEEXISTD10 $\rightarrow$ POLE $\rightarrow$ TDTRI $\rightarrow$ IID $\rightarrow$ DMS;*
- (3) *the same with WEXT, TDLTRI in place of POLE, TDTRI.*

Here BASE12, SIMPLE14, B14D are the finite statements of Section 4 entered as named hypotheses.

Remark 3.13 (What the formal version does not yet cover). (i) BASE12, SIMPLE14, B14D are proved in prose with computer assistance and are *partly* certified in separate Lean modules (Section 4), but these modules are not yet linked to the named hypotheses of Theorem 3.12; in particular the completeness of the graph census lists is established by computation outside Lean. (ii) Statement fidelity—that the Lean propositions mean the prose statements—is proved in Lean for (H), (II),

(II_D) and DMS; for FEEXTD10, FEEXSTD10, POLE, TDTRI it was checked only by the model-based verifiers.

4. FINITE CASES CERTIFIED BY COMPUTATION

4.1. Method. Every finite fact below has the form “every member of an explicit finite family is EX1-good” (or a pole-dominance or leaf-colouring analogue). Certification in *Lean* follows one pattern: the family is represented by an explicit list of edge lists (orbit representatives under automorphisms where applicable); for each representative, edge g and $t \in \{0, 1\}$, the certificate supplies an explicit colouring; a Boolean checker (e.g. a star 6-colouring checker with a neighbour table, and a check that colour class 6 is a PM with the prescribed status of g) is defined in *Lean*; its *soundness* theorem (checker returns **true** $\Rightarrow$ the mathematical property holds) is proved; and the checker is evaluated on the data by the kernel (**decide**), not by compiled code. Hence no trust in external code is needed for the certified statements, but the *completeness* of a census list (that every graph of the class is isomorphic to a listed one) is, unless stated otherwise, a separate computation.

4.2. Results.

Proposition 4.1 (BASE12). *Every $X \in \mathcal{D}$ with $|V(X)| \in \{10, 12\}$ is EX1-good; in fact for every edge g and $t \in \{0, 1\}$ there is a PM N with $[g \in N] = t$ such that (X, N) is MC-colourable. (Lean-certified.)*

Proposition 4.2 (SIMPLE14). *Every simple 3-edge-connected cubic graph on 14 vertices is EX1-good, in the same strong form. (Computer-assisted; Lean certificates exist for all 341 graphs of the census list, in five parts.)*

Proposition 4.3 (B14-D). *Every non-simple $X \in \mathcal{D}$ with 14 vertices is EX1-good, in the same strong form. (Computer-assisted; the census has 984 orbit representatives, of which parts 0–10 out of 14, i.e. representatives $R_0, \dots, R_{791}$, are Lean-certified at the time of writing.)*

Proposition 4.4 (SMALLHOST-D). *Let H be a c4c simple cubic graph with $|V(H)| \leq 8$ and $D \subseteq E(H)$ with $|V(H)| + 2|D| \geq 16$. Then H^D is EX1-good, in the strong form. (Lean-proved, including the reduction to 451 representative markings of the hosts K_4 , $K_{3,3}$, Q_3 and the Wagner graph V_8 , at 1, 13, 129 and 308 markings respectively.)*

Proposition 4.5 (Further finite facts).

- (a) *Every connected bridgeless loopless cubic multigraph on at most 8 vertices has a star edge colouring with 6 colours; for every such Y and every edge g , $T(Y, g)$ has an MC-leaf colouring (a star 6-colouring in which every vertex meets exactly one edge of colour 6).*
- (b) *Every $X \in \mathcal{D}$ with $10 \leq |V(X)| \leq 14$ has an MC-leaf colouring of $T(X, g)$ at every eligible edge g .*
- (c) (SMALL-PD, the instance used) *The pole $Q(Y, z)$ is dominant when Y is the prism, $K_{3,3}$ or V_8 and z is any vertex, and when Y is K_4 with a digon inserted into one edge ab and $z \in \{a, b\}$. (Lean.) A more general computer-assisted version covers the poles of all digon insertions of K_4 , $K_{3,3}$, Q_3 , V_8 with two explicit families of possible exceptions.*

5. PARTIAL RESULTS ON THE OPEN STATEMENTS

Theorem 5.1 (Small cases of the open statements).

- (a) (POLE) *holds for all $X \in \mathcal{D}$ with $|V(X)| \in \{10, 12, 14, 16\}$; in fact the poles admit W -colourings as required by (W-EXT). (Computer-assisted.)*
- (b) (II_D) *holds for all G_0 with $|V(G_0)| \leq 14$.*

Theorem 5.2 (Far-exchange completion under spacing conditions). *Let G be a finite simple cubic graph, M a PM and C a far-exchange set for M (simple form). [For this draft we do not reproduce the notions of path gaps, cyclic gaps, exchanges and gap-distance used below (a cyclic gap is a cycle of $E(G) \setminus M$ containing no edge of C ; path gaps are the corresponding paths); they must be imported for a final version.]*

- (a) *If every endpoint of every exchange lies at gap-distance at least 2 from the ends of its own path gap (with distinct gaps for distinct endpoints, and a small excluded position on gaps of length 4), then G has a star edge colouring c with 6 colours with $c^{-1}(6) = M$ and $c^{-1}(5) = C$.*
- (b) *The same conclusion holds if every exchange endpoint is “deep” and endpoints on a common gap are at distance at least 4 along it.*

These cover the “global” part of (FE-EXT-D)₁₀ in the case $D = \emptyset$: a further reduction shows that far-exchange completion follows from the existence of explicit local data around each exchange, and the remaining difficulty is purely local (Section 8).

6. STAR COLOURINGS OF SPECIAL FAMILIES

For integers $n \geq 5$ and $1 \leq k < n/2$ the generalized Petersen graph $GP(n, k)$ has vertices u_i, v_i ($i \in \mathbb{Z}/n$) and edges $u_i u_{i+1}$, $u_i v_i$ (spokes), $v_i v_{i+k}$.

Theorem 6.1 (Spokes as a colour class). *For every $k \geq 5$, $GP(k, 2)$ has a star edge colouring c with 6 colours such that $c^{-1}(6)$ is exactly the set of spokes. (Lean-proved.)*

Thus these graphs satisfy the “MC” form (a PM as colour class) for a specific, prescribed perfect matching.

Theorem 6.2 (Star 5-edge-colourable families). *Each of the following graphs has a star edge colouring with 5 colours:*

- (a) *the flower snarks J_n , $n \geq 5$ odd;*
- (b) *the Goldberg snarks G_k , $k \geq 5$ odd;*
- (c) *every finite simple graph admitting a covering map to the Petersen graph; every finite simple graph admitting a covering map to K_4 ;*
- (d) *$GP(2m, k)$ for $m \geq 3$ and odd $k < m$; $GP(4m, k)$ for $m \geq 2$ and odd $k < 2m$;*
- (e) *$GP(10m, 4)$, $GP(14m, 6)$, $GP(17m, 8)$, $GP(22m, 10)$, $GP(26m, 12)$, $GP(30m, 14)$ for every $m \geq 1$;*
- (f) *every graph obtained from a loopless cubic multigraph by replacing each vertex by a copy of the Petersen graph minus a vertex (more generally, an “all-good” inflation).*

Moreover star colourings pull back along covering maps: of simple graphs, and of multigraphs along parallel-respecting covers.

Remark 6.3. Two further statements—star 5-colourability of $GP(n, 2)$ for all $n \geq 5$ and of $GP(n, 3)$ for all $n \geq 7$ —were produced and verified by the language-model verifier, but their cross-check audits are *disputed* (the written window-enumeration case split is not exhaustive, although the colourings were independently regenerated and confirmed for $n \leq 399$ in the $GP(n, 3)$ case). We do not claim them here, nor two results that depend on them. Families (c)–(f) of Theorem 6.2 overlap with published work on generalized Petersen graphs (Section 9.4).

7. COMPUTATIONAL EVIDENCE

7.1. Global sweep (imported). The project imported, from earlier (pre-harness) work, a record that every connected loopless graph in the following classes is star 6-edge-colourable, about 30.7 million graphs in total: simple cubic graphs with $n \leq 22$ vertices (7,875,918 graphs, of which 7,319,447 with $n = 22$); simple subcubic graphs with $n \leq 17$ (22,473,106); cubic multigraphs with $n \leq 18$ (376,509); subcubic multigraphs with $n \leq 10$ (7,925). The same record reports exact star

chromatic indices for cubic graphs with $n \leq 18$, and that among bridgeless cores in the swept range only $K_{3,3}$, the prism and two small multigraph gadgets need six colours. *Caveat:* this sweep was not re-run by the harness; its artifact is recorded as an early snapshot, and it has the status “computational evidence”, not “verified fact”.

7.2. Evidence for the open statements. Recorded census results (all “evidence” status, not facts):

- (TD-TRI): exhaustive over its exact scope at $|V(X)| = 16$ and 18 (92,694 instances, 2.79M realizable pairs), a sample at 20 vertices (1.06M pairs) and stress hosts on 12–100 vertices, with 0 misses; an independent census through 20 vertices (≈ 34 M pairs), also 0 misses.
- (FE-EXIST-D)₁₀: exhaustive for $|V(X)| = 16, 18, 20$ (≈ 26 M pairs) and all D on c4c hosts with 10 and 12 vertices; no failure.
- (FE-EXT-D)₁₀: targeted enclosed-exchange structures (14,693 structures, 0 failures); for $D = \emptyset$, local-data constructions succeeded on all tested instances (one exchange: all c4c graphs with $n \in \{10, 14\}$, 17,137 instances, and all 6,100 c4c graphs with $n = 18$, 53,964 instances; two exchanges: all 607 c4c graphs with $n = 16$, 4,801 instances; random $n = 30$), with capped enumerations.
- (II_D): 120 random hosts with 14–28 vertices.

8. OPEN PROBLEMS

By Theorem 3.9, DMS for loopless multigraphs follows from:

Open Problem 8.1 ((FE-EXT-D)₁₀; contract P18, with the case $D = \emptyset$ as P09). Prove Statement 3.7 (FE-EXT-D)₁₀.

For $D = \emptyset$ the global part is reduced to local data (Theorem 5.2 and the reductions after it). The remaining single-exchange local lemma is a joint constraint problem in a bounded zone around the exchange (“ZONE”). Each piece is easy in isolation, but the joint choice under crowding is not: a deterministic greedy fails on 189 of 3,189 instances with $n \leq 14$, although a small search repairs them; and labelled radius-2 neighbourhoods are essentially all distinct, so no finite catalogue will close the lemma. A crowding-tolerant argument is needed. With several exchanges at the minimum allowed distance 3, the local zones interact in most instances, so the multi-exchange case is not a sum of single-exchange problems.

Open Problem 8.2 ((FE-EXIST-D)₁₀; contract P16). Prove Statement 3.7 (FE-EXIST-D)₁₀.

This is a pure existence statement (a PM with prescribed status of g plus an edge set; no colouring). The version over all of $\mathcal{D}$ is false (Remark after Theorem 3.9), so any proof must use cyclic 4-edge-connectivity of the host.

Open Problem 8.3 ((POLE); contract P14). Prove Statement 3.7 (POLE). By Theorem 5.1(a) and Remark 3.10 it suffices to treat members with at least 18 vertices, split into those without a 3-edge-cut with distinct ends and an explicit residue.

A “re-pairing” route was abandoned after its existence hypothesis was found to be false.

Open Problem 8.4 ((TD-TRI); contract P23). Prove Statement 3.7 (TD-TRI).

A one-crossing gluing step for (TD) across a 3-edge-cut with distinct ends is verified, but it only transfers pairs (g, t) realized by a constrained PM crossing the cut exactly once; at 16 vertices 25–30% of the relevant cuts have pairs realized only by PMs crossing three times, so an induction over 3-edge-cuts needs a separate three-crossing step (and a named c4c core statement).

Open Problem 8.5 ((II_D); contract P19). Prove Statement 3.7 (II_D) for hosts with at least 16 vertices.

Open Problem 8.6. Decide whether every connected bridgeless simple cubic graph other than $K_{3,3}$ and the prism is star 5-edge-colourable (Theorem 3.2).

9. STATUS OF VERIFICATION

9.1. How the results were produced. All results were generated by an automated system of language-model worker agents. A result enters the project’s fact graph only after acceptance by a language-model verifier (“llm”). Most facts were additionally re-checked by an independent cross-check audit using a different model (“audit”), which re-reads the proof, checks non-vacuity of hypotheses and the consistency of internal citations; its verdict is *passed* or *disputed*. Facts stated as **Lean** theorems were additionally compiled by a gate that rejects **sorry**, **native_decide** and non-standard axioms (“Lean”). No human has checked the proofs in detail. The human operator only made scheduling decisions and resolved one dispute (below).

9.2. Status of the results in this paper.

Result	Verification
Thm 3.1 ((I)^(II) $\Leftrightarrow$ DMS)	llm + audit; Lean version of $\Rightarrow$
Thm 3.2 (CubicSharp5 _s $\Rightarrow$ DMS)	llm + audit; Lean
Thm 3.3 (HOLE $\Rightarrow$ DMS)	Lean (+ audit)
Thm 3.5, 3.6	llm + audit; Lean complete, with statement fidelity
Thm 3.8, Main Thm 3.9	llm + audit; Lean modulo BASE12, SIMPLE14, B14D; fidelity of hypotheses not formally checked
Prop 4.1, 4.4, 4.5(c)	Lean
Prop 4.2	computer-assisted, llm + audit; Lean parts, 3 of 5 audit-disputed
Prop 4.3	computer-assisted, llm + audit; Lean parts 11 of 14
Prop 4.5(a),(b), Thm 5.1	computer-assisted, llm + audit
Thm 5.2	llm + audit
Thm 6.1	Lean (+ audit)
Thm 6.2	llm + audit (covers: also Lean)
Section 7	evidence only (not facts)

9.3. Open audit disputes. The following facts in the transitive support of the main chain carry a disputed cross-check audit:

- Lemma 2P (MC colourings across a 2-edge-cut): an omitted subcase in a 4-walk case split. The operator marked the dispute resolved: a separate audit-passed lemma covers the omitted subcases, and the formal proof of Theorem 3.6 shows the disputed part is not used on the root path.
- Lemma LEAF-FLIP (a leaf-case lemma): the hand proof was judged sound; the dispute concerns an overstated scope of a computational cross-check cited in the write-up.
- Two **Lean** layers in the import chain of Theorem 3.12: layer 5 (an incorrect claim in the prose outline, not in the **Lean** statement) and layer 19 (two conjuncts have hypotheses now known to be false, hence vacuous; they are not used by the main chain). For kernel-checked theorems these disputes do not affect validity, only the accompanying prose.
- Certificate parts S14-L0, S14-L3, S14-L4 (Proposition 4.2): identification of **Lean** edge lists with the census list.

Disputes in side branches not used by the main theorem: RED-L and RED-Q (reductions for (FE-EXT-D) with $D = \emptyset$), BRICK-TD3 and REQ, KILL- δ , KILL-T, KILL-T5, CORE-FE, several colouring lemmas, and the $GP(n, 2)$, $GP(n, 3)$ results.

9.4. **Novelty.** For most facts the harness’s novelty check was not run (“unchecked”); a minority are marked “new”, “partly known” or “unclear” by a model-based check, which is not a literature review. A light search by the author of this draft found: star 5-edge-colourings of $GP(n, k)$ are known when $\gcd(n, k) \geq 3$ (Zhu et al. [5], completed by Omoomi and Vahid Dastjerdi [6], who also treat some cases with $\gcd = 2$), so parts (d) and (e) of Theorem 6.2 may overlap with [6]; that star colourings pull back along covering maps is elementary and likely folklore; values such as χ'_s of the Petersen graph are classical [1]. We found no reference for the reduction chain of Section 3, for Theorem 6.1, or for the snark families, but a proper literature review is future work.

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The results were produced with an automated proof-search harness using language-model agents and the `Lean` theorem prover. [Operator to complete.]

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